

0:00 — 0:30

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JOURNAL CLUB PRESENTATION

Explainable Artificial Intelligence

Core ideas, techniques, and solutions —
what it is, why it matters, and how to use it in practice

Dwivedi et al., 2023

ACM Computing Surveys

~20 min

CITATIONS	DOWNLOADS	PUBLISHED
986	42,370	Jan 2023

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OVERVIEW

What we'll cover today

PART 1 — 6 MIN

The Problem

Why ML models need explanation. The black box problem. Real consequences of unexplainable AI.

PART 2 — 8 MIN

The Methods

Taxonomy of XAI techniques. SHAP, LIME, and how they work. Global vs. local explanations.

PART 3 — 6 MIN

The Takeaway

What XAI actually adds beyond running SHAP. When it's required. How to choose the right tool.

Central question: If I already know SHAP and LIME, what does "XAI" tell me that's genuinely new?

1:00 — 3:30

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PART 1 — THE PROBLEM

Why accuracy alone isn't enough

A model can be 96% accurate and completely wrong about *why*.

THE CLEVER HANS PROBLEM

A model trained to detect pneumonia from X-rays learned to recognize **hospital ID tags** in the corner — not lung tissue. 96% accurate. Completely useless clinically.

"Just having a highly accurate model is not sufficient to trust and deploy the model in real-world applications."

— Dwivedi et al., 2023

GOOGLE IMAGES, 2015

Searching "CEO" returned women only **11%** of the time despite real-world representation of 27%. The model learned bias from training data — invisibly.

THREE THINGS ACCURACY METRICS MISS

- **What the model actually learned** — could be spurious correlations
- **Whether reasoning is fair** — bias may be invisible in aggregate metrics
- **Why a specific decision was made** — needed for audits and appeals

LOAN DENIAL (GDPR)

EU law requires any automated decision affecting a person to be **explainable on request**. A neural net prediction alone cannot satisfy this — legally.

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PART 1 — THE PROBLEM

Student question: Is XAI just a methodology?

"If I can run a neural network and then run SHAP without knowing XAI — what has XAI told me new?"

1 — UNIFIED VOCABULARY

Before XAI, SHAP, LIME, and saliency maps were scattered across different research communities. XAI packages them with consistent terms: global vs. local, model-agnostic vs. specific, post-hoc vs. intrinsic.

2 — LEGAL OBLIGATION

XAI defines *when* explanation is legally required — not just technically possible. GDPR, healthcare regulation, financial audits. A data scientist running SHAP for curiosity vs. for compliance is a different act.

3 — CATCHES WHAT ACCURACY MISSES

XAI is the only framework that systematically checks whether a model's *reasoning* is valid — not just its output. This is the Clever Hans problem. You can't find it without an explanation tool.

Short answer: XAI is not a new algorithm. If you already use SHAP on your neural network, you are already doing XAI. What XAI adds is the **structured argument that you are *obligated* to do it** in certain domains — and the taxonomy of which tool answers which question.

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PART 1 — THE PROBLEM

When does explainability actually matter?

LOW STAKES · LOW HUMAN INVOLVEMENT

Optional

XAI is optional

Spam filter · Movie recommendation · Ad placement · Search ranking

User gets a bad recommendation — recoverable, low consequence.

HIGH STAKES · HIGH HUMAN INVOLVEMENT

Legally required

XAI is mandatory

Loan approval · Medical diagnosis · Criminal risk scoring · Hiring decisions

GDPR Article 22: right to explanation for automated decisions.

MEDIUM STAKES · AUDITABILITY
NEEDED

Strongly
recommended

XAI for compliance

Fraud detection · Insurance pricing · Predictive maintenance

Model may be audited — need to show it isn't biased against protected groups.

ANY STAKES · DEVELOPER NEEDS

For debugging

XAI for model quality

Any model in development · Human reviews final decision

Catching Clever Hans behavior before deployment — every data scientist's responsibility.

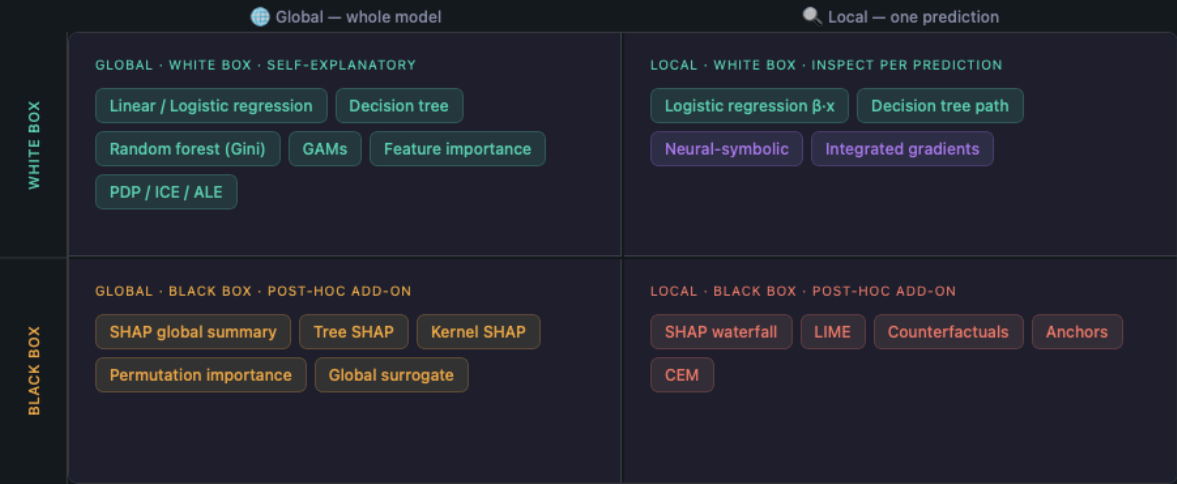
7:00 — 9:00

PART 2 — THE METHODS

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Taxonomy of XAI techniques

Three axes classify every XAI method: scope · model dependency · transparency



Deep learning always lives in the black box rows — no white-box option once hidden layers are added.

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PART 2 — THE METHODS

SHAP — the workhorse of black-box explanation

Game-theoretic: asks "what is each feature's *fair* contribution to this prediction?"

THE CORE IDEA

For each prediction, compute the average marginal contribution of a feature across all possible subsets of features. The result ϕ_i is that feature's "Shapley value" — its fair share of credit for the prediction.

THE ADDITIVE PROPERTY

$$\sum \phi_i = f(x) - E[f(x)]$$

All SHAP values for one prediction always sum exactly to the prediction's deviation from the dataset mean. This is mathematically guaranteed — unlike LIME.

TREE SHAP

Exact values. Fast. Only works on tree-based models (XGBoost, RF).

KERNEL SHAP

Approximate. Works on any model including neural nets. Slower.

LOCAL SHAP — LOAN DENIAL EXAMPLE

baseline (avg approval) = 0.50



Final prediction (denial probability) **0.87**

▲ pushes toward DENIED · ▼ pushes toward APPROVED

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PART 2 — THE METHODS

SHAP global feature importance

Aggregate local explanations across the whole dataset to understand overall model behavior.

HOW IT WORKS

- **Step 1:** Compute local SHAP values for every row → $N \times F$ matrix of ϕ_i values
- **Step 2:** Take mean $|\phi_i|$ per feature across all rows
- **Step 3:** Rank features by that mean absolute value

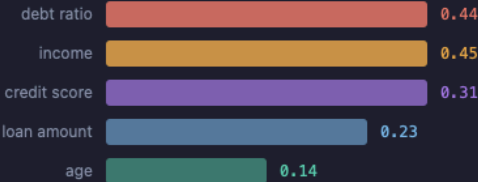
WHY ABSOLUTE VALUE MATTERS

Income pushes some predictions up (+0.73) and others down (−0.18) — but its average $|\phi| = 0.45$ still ranks it as an important feature. Signs cancel; magnitude is preserved.

SHAP VS. SKLEARN FEATURE_IMPORTANCES_

Built-in Gini importance is **biased toward high-cardinality features**. Tree SHAP gives unbiased, exact values. Always prefer SHAP for final reporting.

GLOBAL SHAP — MEAN $|\phi_i|$ RANKING



Beeswarm plot adds color per feature value → shows direction + distribution simultaneously

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PART 2 — THE METHODS

LIME — local approximation of any black box

Explains one prediction by building a simple interpretable model in its local neighborhood.

- 1

Take one data point to explain

The specific instance whose prediction you want to understand — e.g. one loan applicant.
- 2

Perturb the input hundreds of times

Slightly modify feature values, run each through the black-box model, collect predictions.
- 3

Fit a local linear model on those samples

Weight samples by proximity to the original point. Fit a simple linear regression to approximate behavior locally.
- 4

Read off the linear coefficients as explanation

Coefficients of the local linear model = feature contributions for that one prediction.

LIME VS. SHAP — WHEN TO USE WHICH

	LIME	SHAP
Speed	Faster	Slower (Kernel)
Stability	Can vary between runs	Consistent
Additivity	Not guaranteed	Always sums to $f(x)$ $-E[f(x)]$
Best for	Quick exploration	Final reporting

Key limitation: LIME defines "neighborhood" arbitrarily. Two runs on the same instance can give different feature weights. Use SHAP for anything you're going to show a stakeholder.

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PART 3 — THE TAKEAWAY

Student question: which method do I run and when?

"After running a neural network, to answer the question of reasoning — SHAP, LIME, or something else?"

YOUR MODEL	YOUR QUESTION	USE THIS	WHY
Neural net	Which features matter overall?	Kernel SHAP — global	No built-in importance; SHAP is model-agnostic
Neural net	Why was THIS prediction made?	SHAP waterfall	Additive, stable, auditable — use for stakeholders
Neural net	Quick exploration / what-if?	LIME	Faster; unstable so don't use for final reporting
XGBoost / RF	Which features matter overall?	Tree SHAP — global	Exact, fast — do NOT use sklearn .feature_importances_
XGBoost / RF	Why was THIS prediction made?	Tree SHAP — local waterfall	Exact Shapley values, polynomial time on trees
Logistic reg.	Which features matter overall?	Coefficients directly	White box — β IS the explanation. No add-on needed.
Logistic reg.	Why was THIS prediction made?	$\beta \cdot x$ per instance	Multiply coefficient $\times$ feature value per row. Exact.
Any model	Is the model biased / fair?	SHAP beeswarm	Check if protected attributes rank high in mean $ \phi_i $

Rule of thumb: SHAP for anything that goes to a stakeholder or audit. LIME for quick interactive exploration. Coefficients if you're still on a white-box model.

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PART 3 — THE TAKEAWAY

What this paper actually contributes

NOT A NEW ALGORITHM

XAI does not invent SHAP, LIME, or PDP. These already existed. XAI is the framework that organizes them, names the problem they solve, and defines when you are **obligated** to use them.

"Model performance metrics do not always exhibit true prediction decisions. Just having a highly accurate model is not sufficient to trust and deploy the model."

— Dwivedi et al., 2023

A TAXONOMY YOU CAN ACTUALLY USE

Global vs. local, white box vs. black box, model-specific vs. agnostic. These three axes let you navigate from "I have this model and this question" to "I should use this tool" without guessing.

THE THREE THINGS XAI GIVES A DATA SCIENTIST

- **Debugging:** find Clever Hans behavior before deployment
- **Compliance:** satisfy GDPR and domain-specific audit requirements
- **Communication:** explain model decisions to non-technical stakeholders, regulators, and affected users

ACCURACY ≠ TRUSTWORTHINESS

The paper's central empirical argument: high accuracy is a necessary but not sufficient condition for deployment. XAI provides the second check — is the *reasoning* valid?

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DISCUSSION — 20:00 ONWARDS

Open questions for the group

Q1 XAI is described as a "methodology, not a new algorithm." Do you think that framing understates or overstates its contribution to the field?

Q2 SHAP values tell you which features drove a prediction — but they don't tell you if those features are ethically appropriate to use. Where does XAI end and ethical AI begin?

Q3 There is a known accuracy–interpretability tradeoff. Should regulators mandate explainability even if it means deploying a less accurate model in healthcare or finance?

Q4 With LLMs now making decisions in real applications, do you think SHAP and LIME are sufficient XAI tools — or does the field need entirely new explanation methods for generative AI?

Dwivedi et al. (2023) · ACM Computing Surveys · doi:10.1145/3561048

Thank you